Experiment 7 - Bragg's Law

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1 Background

In 1912 German physicist Max von Laue discovered that when X-rays passed through a crystal sample diffraction and interferance occured. The effect was observed when a crystal was struck with a broad-band X-ray source and manifested as a ring-like pattern of dots on a photographic plate placed behind the crystal. As von Laue was investigating if X-rays were particles or waves, physicists William H. Bragg and William L.

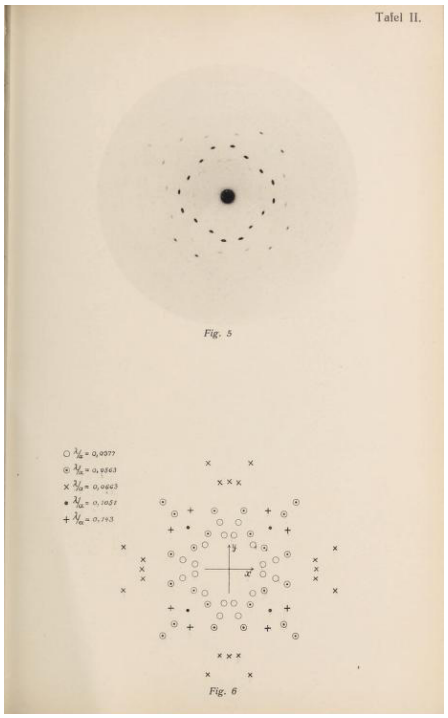


Figure 1: Interferance pattern from X-rays passing through a crystal sample

Bragg investigated if the crystals could be used to reflect the X-rays.

Using an x-ray source, an adjustable apature and a turntable they were able to determine that scattering only occured when there was constructive interferance between the x-rays. They proposed that

$$n\lambda = 2d\sin\theta \tag{1}$$

was the condition for the reflected x-rays to be detected.

More recently,

2 References

1. William Henry Bragg, William Lawrence Bragg, 1913, *The reflection of X-rays by crystals*, Proc. R. Soc. A1 July 1913

2. W. Friedrich, P. Knipping, M. Laue, presented by A. Sommerfeld 1912, *Interference Phenomena for X-Rays, Theoretical section*, Proceedings of the Mathematical-Physical Class of the Royal Bavarian Academy of Sciences in Munich, Issue 2. English translation by J. J. Stezowski.